Martin Harris

Senior Backend Engineer · Distributed Systems · Edinburgh, UK cv@nakomis.com · +44 1312 125 959 · cv.nakomis.com · github.com/nakomis

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Professional Summary

Senior backend engineer with deep experience building and operating distributed systems across cloud-native and on-premises environments. Primary backend stack is JVM (Java, Kotlin), with extensive TypeScript across frontend, IaC, and serverless surfaces. Strong background in AWS, cloud orchestration, and event-driven architectures. Recent work spans regulated financial-services platforms and large consumer-facing loyalty systems. Apache Software Foundation committer (Apache Brooklyn).

Skills

Backend & Distributed Systems: Microservices, event-driven architecture, REST APIs, gRPC, OIDC/OAuth, cloud orchestration, message-driven systems

Languages: Java, Kotlin, TypeScript / Node.js, Python, Go, Bash

Cloud & Infrastructure: AWS (CDK, EKS, Lambda, API Gateway, EventBridge, DynamoDB, S3, CloudFront, IoT Core, Cognito, SageMaker), Google Cloud, Microsoft Azure, Terraform, CloudFormation, Kubernetes, Docker

Frameworks & Tools: Spring Boot, Next.js, React, Jenkins, Drone CI, Datadog

AI & Machine Learning: LLM APIs (Anthropic, OpenAI, Google), AWS Bedrock, AWS SageMaker (custom training pipelines, serverless inference), TensorFlow / MobileNet fine-tuning

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, MS SQL Server

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Professional Experience

Senior Software Engineer

Test Driven Solutions Ltd · Edinburgh, UK · *November 2023 – Present*

Consulting engineer delivering core backend systems for enterprise clients across financial services, consumer rewards, media, and construction services.

- **Cloud-native clearing-and-custody platform** *(January 2026 – Present)* — Backend engineer on a regulated financial-infrastructure platform serving 900+ institutional clients (hedge funds, asset managers, broker-dealers) that collectively reach over 40 million end users globally. Building Java/gRPC microservices behind a TypeScript (Next.js) front end on Kubernetes with PostgreSQL and Datadog. Shipped a real-time chat interface backed by those services for order placement, position management, and market-data / news retrieval, alongside multi-leg options strategy tooling and on-demand market commentary. Stack: Java, Spring Boot, gRPC, TypeScript (Next.js), Kubernetes, Datadog, PostgreSQL.
- **Consumer loyalty and rewards platform** *(October 2024 – December 2025)* — Backend engineer on a unified loyalty platform reaching roughly 35 million customers across

a multinational consumer-brand group. Designed and built a third-party API integration framework that consolidated multiple partner integrations behind a single contract, including a booking.com v3 endpoint for transaction-based earning. Delivered a marketing-campaign system on AWS EventBridge and an activity-based earn engine. Stack: Java, Spring Boot, Python, TypeScript, AWS (EventBridge, Lambda, RDS), Terraform, Drone CI.

- **Fire protection and construction services company** *(June 2024 – October 2024)* — Migrated the client's Android build pipeline and Play Store hosting in-house for a UK-wide field workforce of 100+ engineers, and updated the Kotlin codebase for compliance. Cut handset-to-server sync time from 5–10 minutes to around 20 seconds by collapsing 1,000+ per-record HTTPS calls into a single batched request. Stack: Kotlin, Android, VB.NET, SQL Server.
- **Major UK public broadcaster** *(December 2023 – June 2024)* — Contributed to the identity-platform migration (IDMv1 → IDMv2) underpinning tens of millions of UK accounts, in an agile squad focused on GDPR-compliant under-13 registration flows. Designed and built a reusable WebSocket-based primitive that replaced long-polling for "wait for an external event" flows across the platform; email verification was the first consumer, and the primitive was merged for reuse across the rest of the roadmap. Stack: Java, Spring Boot, AWS CDK, DynamoDB, PostgreSQL, WebSockets API Gateway.

Lead Software Engineer

Cloudsoft Corp Ltd · Edinburgh, UK · *2012 – 2023*

- Lead engineer for cloud orchestration software, with several hundred commits to **Apache Brooklyn** (Apache Software Foundation committer) — an open-source framework for modelling, deploying, and managing distributed applications across cloud and on-premises environments.
- Architected and delivered core-product cloud orchestration software and customer-specific integrations via Cloudsoft's AMP platform, working directly with enterprise clients on deployments across AWS, Azure, Google Cloud, and on-premises infrastructure.
- Worked across Java, JavaScript (React), Python, Go, Docker, Kubernetes, Terraform, MySQL, and MongoDB; mentored a team of three engineers.

Senior Product Designer / Programmer

Supply Chain Systems / AgriWise · Norwich / California · *May 2004 – March 2012*

- Designed and built a .NET software platform for real-time tracking and reporting of supply chain events across fresh-produce supply chains in the UK and the Americas.
- Owned the full architecture of the AgriWise platform, integrating bespoke handheld data-capture devices at farms and packhouses with a centralised reporting system.
- Relocated to California in 2006 to lead US market research and adapt the platform for grower-based organisations on the West Coast.

Education

University of East Anglia, Norwich · *1994–1996*

Certificate of Higher Education — Mathematics

Open Source & Projects

Apache Brooklyn

Apache Software Foundation committer on Apache Brooklyn, an open-source framework for modelling, deploying, and managing distributed applications across cloud and on-premises environments. Multi-year contributor.

nakom.is

Personal website, URL shortener, and technical portfolio built on AWS CDK: serverless AI chat (Anthropic Claude with tool use), automated PDF CV generation (Chromium on Lambda), SSE streaming, React SPA, and multi-domain CloudFront distribution. Open source under CC0. github.com/nakomis/nakom.is

BootBoots

A home cat-monitoring system built on a custom-designed PCB and 3D-printed enclosure, with firmware in embedded C++. Uses a MobileNet v2 model fine-tuned on labelled cat-camera footage via a custom SageMaker training pipeline (not an off-the-shelf model), deployed to a SageMaker serverless inference endpoint. A spray deterrent fires when the neighbour's cat is detected. Connectivity via Bluetooth and MQTT to AWS IoT Core; mobile app authentication via AWS Cognito. Infrastructure managed with AWS CDK.

Nationality: British · Last updated: May 2026